SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)
(Established under Section 3 of the UGC Act, 1956) | Re-accredited by NAAC with 'A++' grade | Awarded Category – I by UGC

Department of Computer Science and Engineering
CA2: Midterm Review(Seminar)
of Industry Internship
Session: 2025-26

Title of the Project:- AI Engineering Internship: Langchain and Document Intelligence Systems

Presented By:
Name: Sharvit Kashikar
PRN: 22070521143

Industry Mentor:
Name: Karthik Kalyan Gudipati
Designation: Principle Product Manager

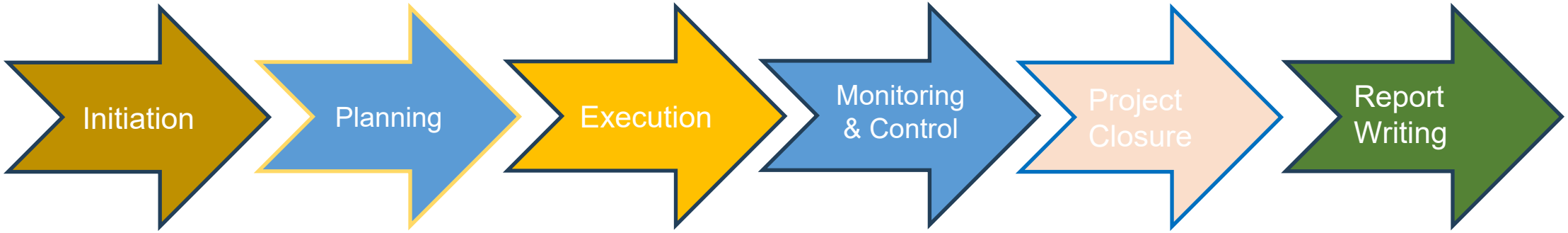
Institute Mentor:
Name: Dr. Amit Bhattacharya
Designation: Assistant Proffesor

Presentation Outline:

SYMBIOSIS
INSTITUTE OF TECHNOLOGY, NAGPUR

- Project Management Timeline
- Suggestions given by mentor in Seminar 1
- About the Company
- Project Description
- Hardware/Software Specification
- Methodology
- Module Description
- Implementation Results (If Available)
- Attendance/Photos of Industry

Project Management Timeline



Task	Initiation	Planning	Execution	Monitoring & Control	Project Closure	Report Writing
Status	Complete	Complete	Complete	Complete	Not complete	Not complete
Remark if any	Internship onboarding & domain training completed	Training roadmap and project objectives defined	Phase 1 and Phase 2 training completed successfully	Weekly progress reviews and mentor feedback sessions conducted	Internship currently in progress	Weekly reports and midterm presentation being prepared

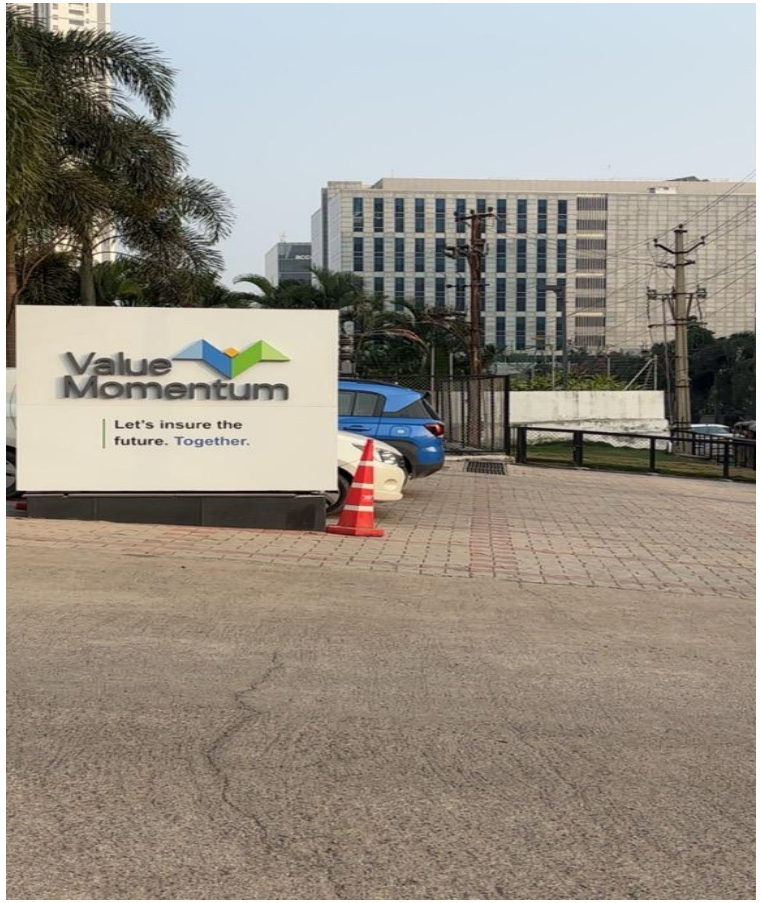
Suggestions Given by Mentor in Seminar 1

Suggestion	Action Taken
Focus on understanding Langchain architecture	Completed the exercises and implemented chains
Improve python architecture understanding	Built smart document processor mini project
Develop LLM applications	Built RAG based document Q&A system

About the Company

- ValueMomentum Software Services Pvt. Ltd. is a global technology services company specializing in digital transformation solutions, with a strong focus on the Property & Casualty (P&C) insurance industry. The company provides services in software development, artificial intelligence, cloud computing, data analytics, and enterprise application modernization across insurance, healthcare, and financial services domains.

- During this internship, I am working with the **Emerging Technologies team**, where I am learning AI engineering concepts such as Python-based AI pipelines, LangChain frameworks, large language model applications, and intelligent document processing systems.



Project Description

Objective:

- To develop practical Skills in AI Engineering using Python and LLM frameworks.
- To build intelligent document processing systems using langchain and AI pipelines.
- To understand modern AI architectures such as transformers, embeddings and agent workflows.

Scope :

- Python-based AI pipelines.
- Langchain-based LLM workflows.
- Retrieval Augmented Generation(RAG).
- Agent based automation.
- Intelligent Document Processing.

Benefits:

- Automates document analysis and information retrieval.
- Improves efficiency in extracting knowledge from documents.
- Enables scalable AI-driven workflows.

Abstract

- This Training phases (phase 1 and 2) focused on developing practical expertise in AI engineering and large language model applications. The training included advanced Python programming, asynchronous pipelines, API integration, and data processing techniques.
- Using the LangChain framework, I learned how to design prompt templates, build modular chains, implement memory systems, and develop custom tools for agent-based workflows.
- As part of the training deliverables, I developed a **Smart Document Processor** using Python and implemented a **RAG-based question-answering system** using LangChain. These projects demonstrate how AI-powered systems process documents, extract insights, and answer queries efficiently using large language models.
- Currently, I am continuing the training by learning advanced concepts such as **LangGraph and multi-agent workflows**.

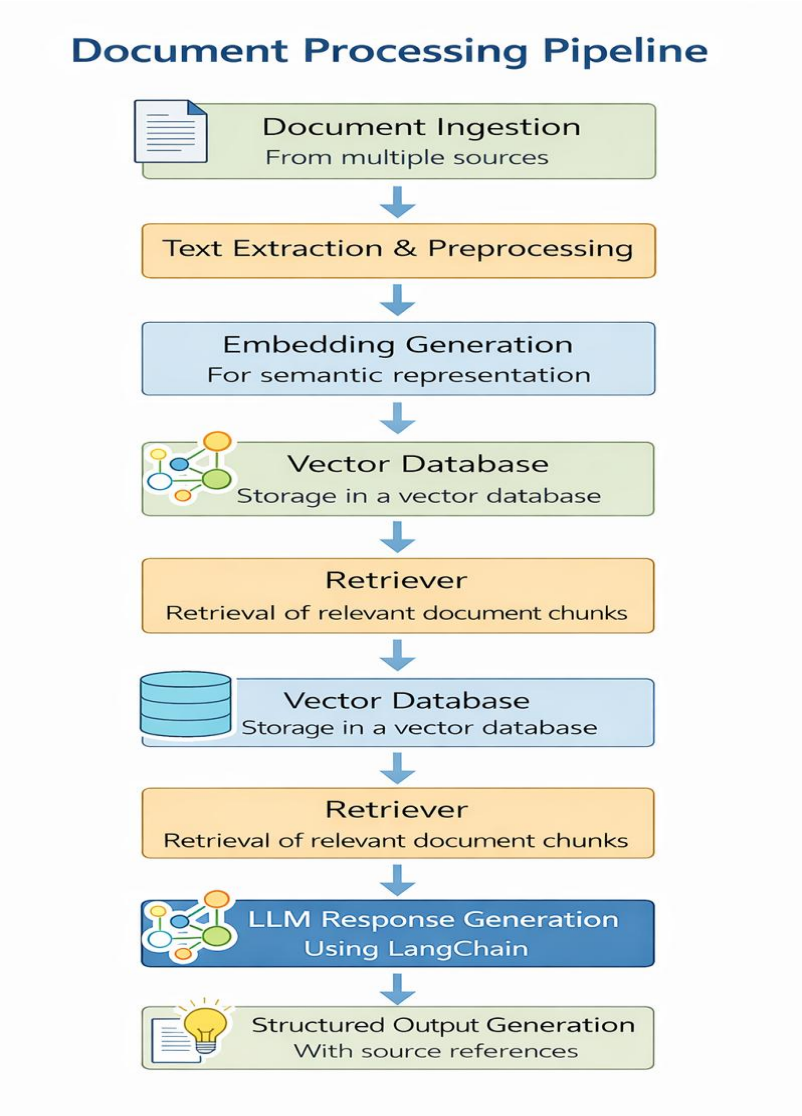
Software Specification

List of Software used:

- Python 3.12
- VS Code
- Git & GitHub
- LangChain
- LangGraph
- OpenAI API
- Pydantic
- Httpx – Asynchronous Requests
- pytest
- FAISS / Vector database (for RAG)
- dotenv

Methodology

- The system architecture for the AI document processing pipeline follows these steps:
- Document ingestion from multiple sources
- Text extraction and preprocessing
- Embedding generation for semantic representation
- Storage in a vector database
- Retrieval of relevant document chunks
- LLM-based response generation using LangChain
- Structured output generation with source references



Module Description

- **Document Processing Module:** This module reads documents in different formats such as **.txt, .pdf, and .docx** and extracts textual information along with metadata.
- **AI Processing Module:** This module performs prompt engineering, interacts with large language models, and processes structured outputs using Pydantic.
- **Retrieval Module:** This module performs document chunking, generates embeddings, and retrieves relevant document segments using vector similarity search.
- **Agent Tool Module:** This module implements custom tools using LangChain and defines structured inputs using Pydantic models.
- **Output Module:** This module generates structured responses and provides relevant source references for the generated answers.

Implementation Results (If Available)

Smart Document Processor

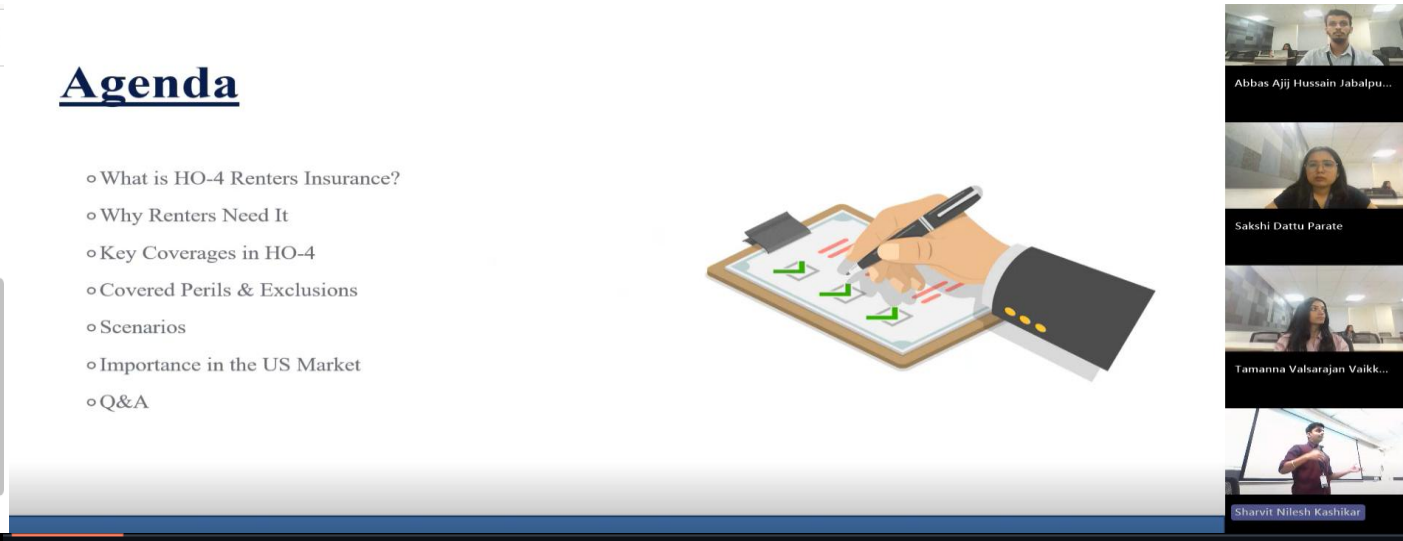
- I developed a CLI-based document processing system.
- The system processes documents asynchronously using asyncio.
- Inputs and outputs are validated using Pydantic models.
- Logging and testing were implemented using pytest.

LangChain RAG System

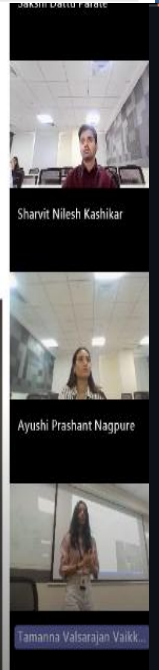
- I configured large language model providers.
- I implemented prompt engineering techniques.
- I built modular pipelines using LangChain Expression Language (LCEL).
- I implemented conversational memory for context management.
- I created custom tools for agent-based workflows.

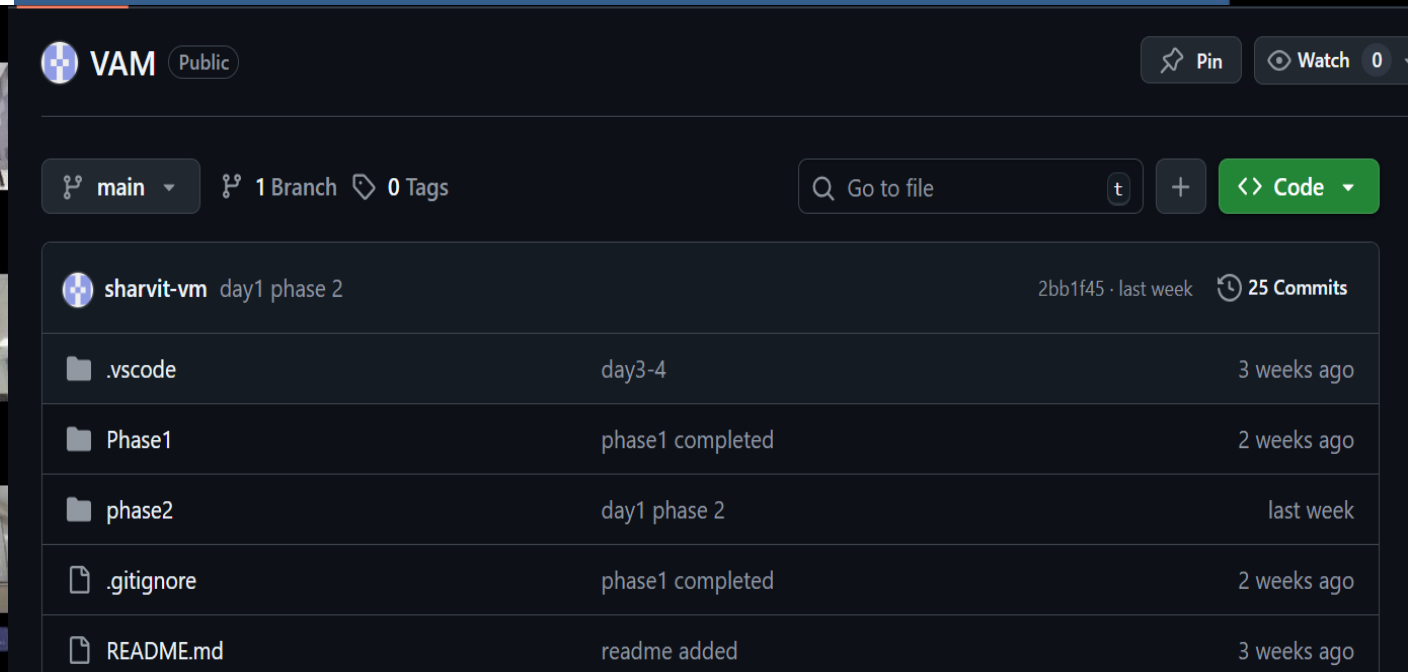
Photos During Internship

Clock In Time	Clock Out Time	Actual Hours	Total Hours	Status	Full day/Half day
				Week off	Full day
09/03/2026 10:23:00	09/03/2026 17:24:00	5:32	7:00	Work from office	Full day
10/03/2026 10:13:00	10/03/2026 17:11:00	6:02	6:58	Work from office	Full day
11/03/2026 10:22:00	11/03/2026 17:29:00	5:34	7:07	Work from office	Full day
12/03/2026 10:22:00	12/03/2026 16:50:00	5:38	6:27	Work from office	Full day
				Remote Attendance Requests	Full day
				Week off	Full day
				Week off	Full day



Dwelling (Coverage A) – The structure of the home
Other Structures (Coverage B) – Garage, fences, sheds
Personal Property (Coverage C) – Furniture, appliances, clothes
Loss of Use (Coverage D) – Living expenses if home becomes uninhabitable
Personal Liability (Coverage E) – Legal responsibility for injuries/damage
Medical Payments (Coverage F) – Minor medical costs for injured guests





Thank You…..